

NARX case (B)

$$\hat{y}(k|\hat{\theta}) = \sum_{i=1}^L \hat{\alpha}_i g[\varphi(k), \tau_i] = \hat{y}(\varphi(k), \hat{\theta})$$

non lin.
basis funct.

"internal"
parameters.

In this case the model is not lin. in $\hat{\theta}$.

⇓
one shot solution is not available. $\Rightarrow$ we have to
resort to math. progr. and need $\frac{\partial J_N(\hat{\theta})}{\partial \hat{\theta}}$

We can use the "chain rule" for partial derivation:

$$\frac{\partial J_N(\hat{\theta})}{\partial \hat{\theta}} = \frac{\partial}{\partial \hat{\theta}} \frac{1}{N} \sum \underbrace{[y(k) - \hat{y}(k|\hat{\theta})]^2}_{e(k|\hat{\theta})} \Rightarrow \frac{\partial e(k|\hat{\theta})}{\partial \hat{\theta}}$$

(B) $\hat{\theta} =_{col} [\alpha_1, \tau_1, \alpha_2, \tau_2, \dots, \alpha_L, \tau_L]^*$

$$\frac{\partial e(k|\hat{\theta})}{\partial \hat{\theta}} = - \frac{\partial \hat{y}(\varphi(k), \hat{\theta})}{\partial \hat{\theta}} \quad \text{includes} \quad \frac{\partial \hat{y}(\varphi(k), \hat{\theta})}{\partial \alpha_i} \dots$$

(135)

$$\dots \frac{\partial \hat{y}(\underline{y}(k), \underline{\hat{\theta}})}{\partial \tau_i} \quad (\text{I 136})$$

$$i = 1, \dots, L$$

$$\left. \frac{\partial \hat{y}(\underline{y}(k), \underline{\hat{\theta}})}{\partial \tau_i} \right|_{\underline{\hat{\theta}} = \underline{\hat{\theta}}^{(e)}} = g[\underline{y}(k), \underline{\hat{\tau}}_i^{(e)}]$$

↑
from
dataset

$$\left. \frac{\partial \hat{y}(\underline{y}(k), \underline{\hat{\theta}})}{\partial \tau_i} \right|_{\underline{\hat{\theta}} = \underline{\hat{\theta}}^{(e)}} = \hat{x}_i^{(e)} \left. \frac{\partial g[\underline{y}(k), \underline{\hat{\tau}}_i]}{\partial \hat{\tau}_i} \right|_{\hat{\tau}_i = \hat{\tau}_i^{(e)}}$$

where $\hat{x}_i^{(e)}$ and $\hat{\tau}_i^{(e)}$ are components of

the parameter vector $\underline{\hat{\theta}}^{(e)} = \begin{bmatrix} \hat{x}_1^{(e)} \hat{\tau}_1^{(e)} & \dots & \hat{x}_L^{(e)} \hat{\tau}_L^{(e)} \end{bmatrix}$

obtained at the

l -th iteration of the
math. progr. alg.

VALIDATION of models / COMPLEXITY of MODELS

(I 137)

Validation of a model corresponds to checking if a model is satisfactory.

This also has to do with comparison of different models.

A good model should have a good performance. Perf. changes with "realizations" $\Rightarrow$ we want to minimize an EXPECTED ~~cost~~ ERROR.

In the case of ~~J137~~ squared error $\Rightarrow$

$$e^2(k|\hat{\theta}) = (y(k) - \hat{y}(k|\hat{\theta}))^2$$

$\Rightarrow$ best $\hat{\theta}$ should be such that

$$\min_{\hat{\theta}} E [y(k) - \hat{y}(k|\hat{\theta})]^2$$

Expect. taken over all poss. realizations.

In general we have a limited data set

$$\Rightarrow \bar{E} e^2(k|\underline{\hat{\theta}}) \rightsquigarrow \frac{1}{N} \sum_{k=0}^N e^2(k|\underline{\hat{\theta}}) = J_N(\underline{\hat{\theta}})$$

exp. value
 $\equiv$ probabilistic
 mean val

Empirical cost
 (using a limited D.S.)

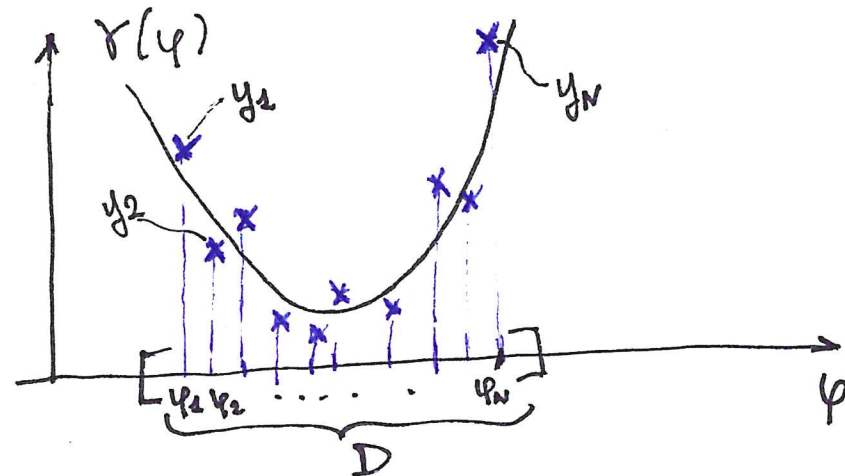
If $N \rightarrow \infty$ we hope

$$J_N(\underline{\hat{\theta}}) \xrightarrow{N \rightarrow \infty} \bar{E} e^2(k|\underline{\hat{\theta}})$$

How to compare two models?

Consider a "regression" problem $\equiv$ approximation of an unknown function using noisy measurement.

Scalar case



Assume we want to approximate the function $\gamma(\cdot)$ in some domain D

Suppose we want to use polynomials to approx the function. (I139)
 Consider 3 possible polyn.

$$\hat{y}_A(\varphi) = \alpha_{A0}$$

$$\hat{y}_B(\varphi) = \alpha_{B0} + \alpha_{B1} \varphi + \alpha_{B2} \varphi^2$$

$$\hat{y}_C(\varphi) = \alpha_{C0} + \alpha_{C1} \varphi + \dots \dots \dots \alpha_{C40} \varphi^{40}$$

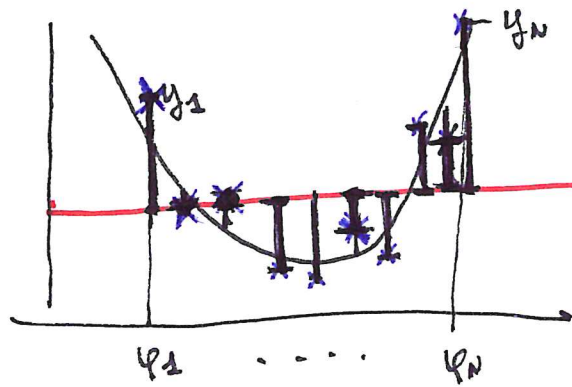
$$\hat{y}_A(\varphi) \longrightarrow \hat{y}(\varphi, \hat{\underline{\theta}}_A) \quad \hat{\underline{\theta}}_A = \hat{\alpha}_{A0}$$

$$\hat{y}_B(\varphi) \longrightarrow \hat{y}(\varphi, \hat{\underline{\theta}}_B) \quad \hat{\underline{\theta}}_B = [\hat{\alpha}_{B0}, \hat{\alpha}_{B1}, \hat{\alpha}_{B2}]^T$$

$$\hat{y}_C(\varphi) \longrightarrow \hat{y}(\varphi, \hat{\underline{\theta}}_C) \quad \hat{\underline{\theta}}_C = [\hat{\alpha}_{C0} \dots \dots \dots \hat{\alpha}_{C40}]^T$$

If we use more parameters we can approximate more "complex" functions. But the function y is unknown, we only have data (measurements)

Using measurements: Data set = $\{\varphi_1, y_1 \quad \varphi_2, y_2 \dots \varphi_N, y_N\}$ (I140)
 with a least squares approach.



Case A:

$$\hat{\alpha}_{A0}^* = \hat{\gamma}(\varphi, \hat{\theta}_A^*) \quad \hat{\alpha}_{A0}^* = \arg \min \frac{1}{N} \sum_{k=1}^N (y_k - \hat{\alpha}_{A0})^2$$

We can do the same for functions

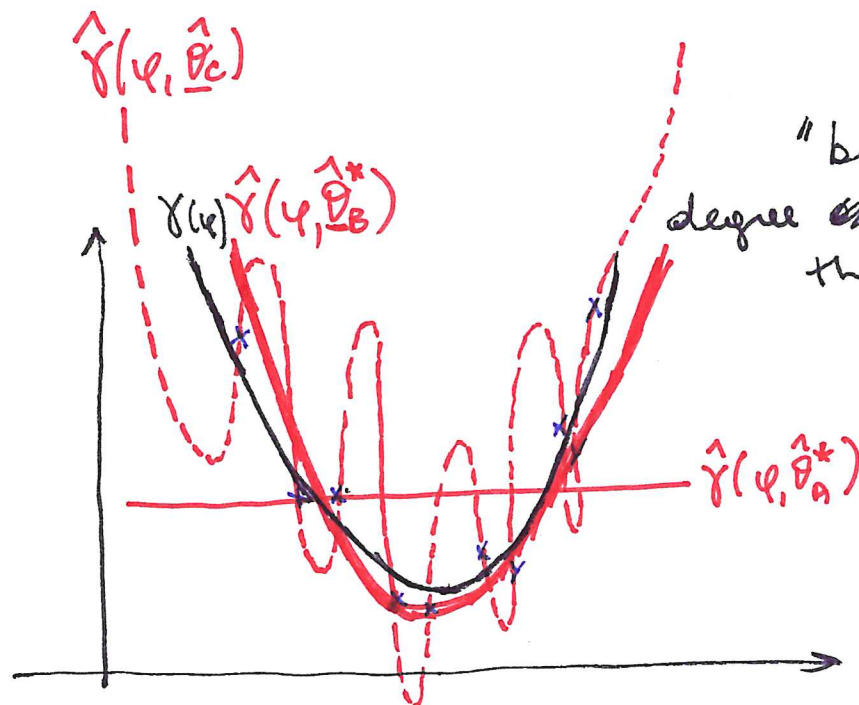
$$\hat{\gamma}(\varphi, \hat{\theta}_B) \quad , \quad \hat{\gamma}(\varphi, \hat{\theta}_C)$$

$$\downarrow$$

$$\hat{\gamma}(\varphi, \hat{\theta}_B^*)$$

$$\downarrow$$

$$\hat{\gamma}(\varphi, \hat{\theta}_C^*)$$



"best poly. of
 degree ~~ex. 2~~ 2 w.r.t.
 the least sq. cost
 with N samples"

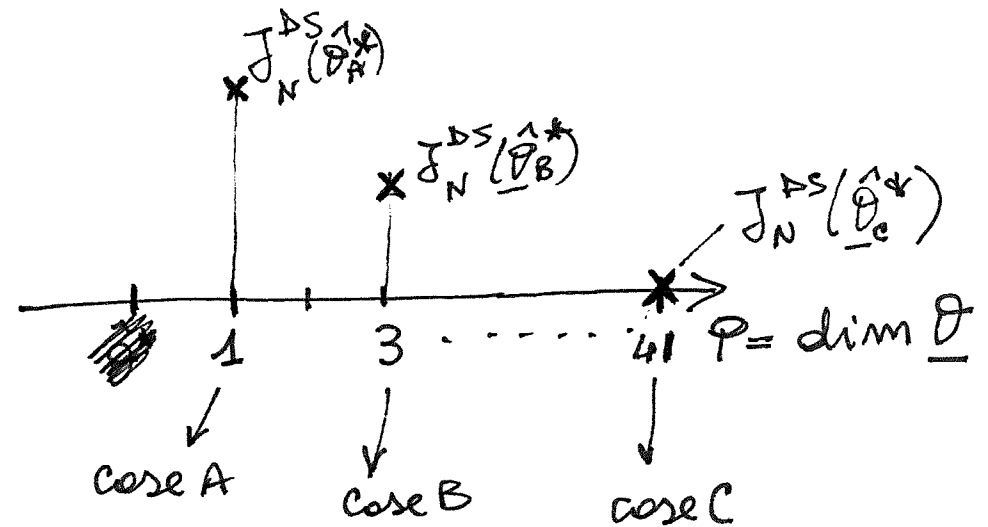
same for
 pol. of degree 40

Best approx. of $\gamma(\varphi)$ is
 given by $\hat{\gamma}(\varphi, \hat{\theta}_B^*)$

BUT The cost for case C here is 0. The points in the data set are perfectly copied.

(I will)

$$J_N^{DS}(\hat{\underline{\theta}}_c^*) = 0$$



As I augment the "complexity" $\Rightarrow J_N^{DS}(\hat{\underline{\theta}}^*)$ monotonically reduces.

BUT $\int_{\varphi \in D} \frac{1}{n} [Y(\varphi) - \hat{Y}(\varphi, \hat{\underline{\theta}})]^2 \cdot d\varphi$ is not minimized by case C

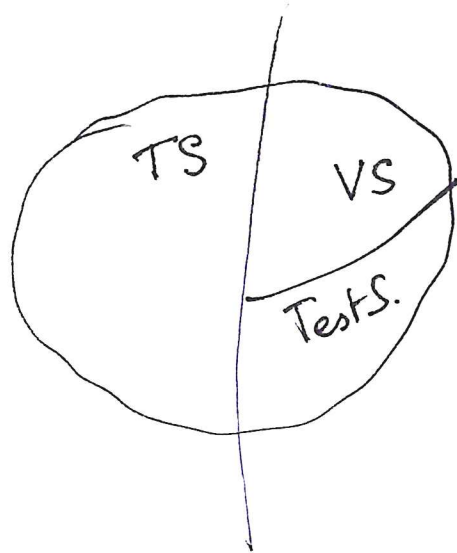
$\underbrace{\hspace{10em}}_{\equiv E e^2 \text{ with const. p.d.f.}}$

⇒ idea : CROSS-VALIDATION.

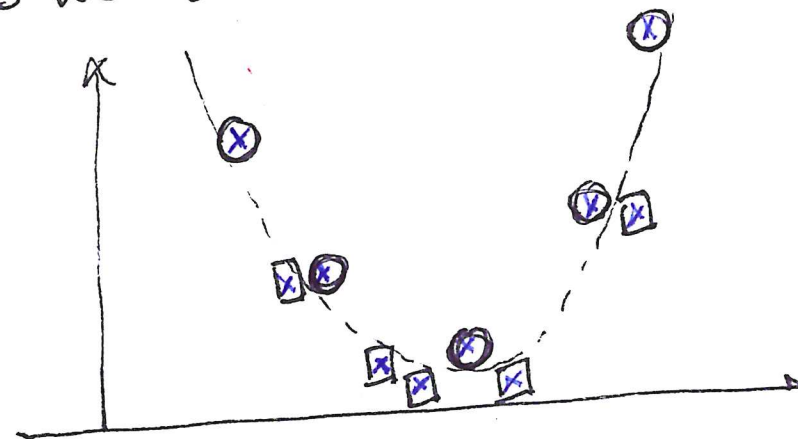
(I u2)

Data Set could be divided in two ^{or} (3) subsets
distinct.

DS. → TS = Training Set
→ VS = Validation Set
→ Test Set

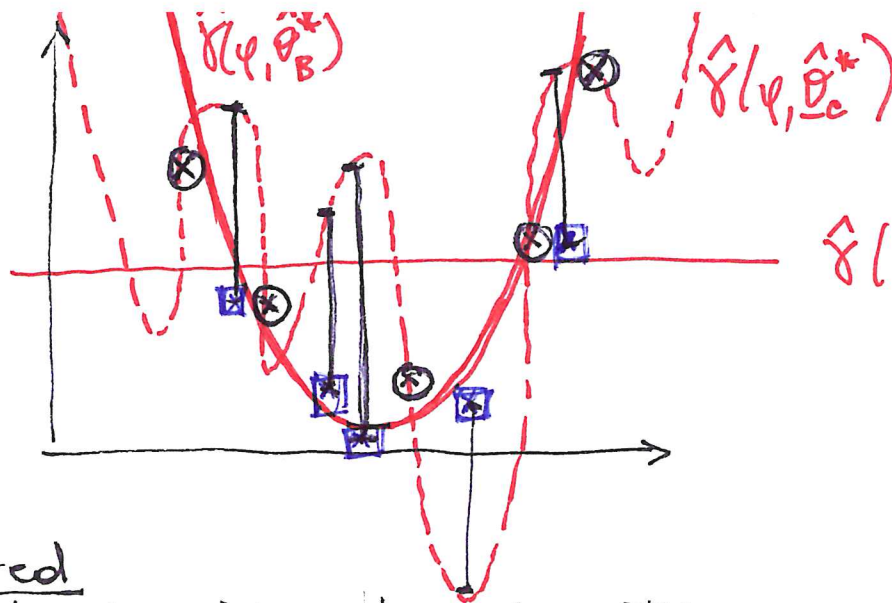


Do not consider Test Set



○ Training Set
□ Validation Set

We will use TS for optimization (~~and~~ Training) of the model
and VS for validating the results.



$$\hat{f}(\psi, \hat{\theta}_A^*)$$

where $\hat{\theta}_A^* = \arg \min_{\hat{\theta}} J^{TS}(\hat{\theta}_A)$

$\hat{\theta}_{A,B,C}^*$ are optimized ~~computed~~ minimizing the cost on TS.

$$J_{(\hat{\theta})}^{TS} = \frac{1}{N_{TS}} \sum_{k \in TS} [y_k - \hat{f}(\psi_k, \hat{\theta}_A^*)]^2$$

cardinality of TS

If I compute the cost on the valid. set without changing $\hat{\theta}_{A,B,C}^*$

Similarly for $\hat{\theta}_B^*, \hat{\theta}_C^*$

$$\text{e.g. } J^{VS}(\hat{\theta}_B^*) = \frac{1}{N_{VS}} \sum_{k \in VS} (y_k - \hat{f}(\psi_k, \hat{\theta}_B^*))^2$$

cardim. of VS

Case C corresponds to "OVERFITTING"
(using too-many parameters)

